Interface Configuration

```
! Configure GigabitEthernet interfaces
Router(config)# interface g0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown

! Configure Serial interfaces
Router(config)# interface s0/0/0
Router(config-if)# ip address 10.0.0.1 255.255.255.252
Router(config-if)# no shutdown
```

RIP Configuration

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# no auto-summary
Router(config-router)# network 192.168.1.0
Router(config-router)# network 10.0.0.0
Router(config-router)# exit
```

Subnet Calculation Reference

Goal: Example of subnet allocation for different departments based on device requirements.

Department	Usable Hosts	Network/ Mask	Subnet Mask	Usable IP Range
Engineering	45	192.28.3.0 /26	255.255.255.192	.1 – .62
Sales	32	192.28.3.64 /26	255.255.255.192	.65 – .126
Marketing	26	192.28.3.128 /27	255.255.255.224	.129 – .158
Admin	14	192.28.3.160 /28	255.255.255.240	.161 – .174
Management	7	192.28.3.176 /28	255.255.255.240	.177 – .190

DHCP Physical Server Config

1. DHCP Server Setup

Assign a static IP address to the DHCP server before creating address pools.

Parameter	Value
IP Address	`192.168.1.2`
Default Gateway	`192.168.1.1`

2. DHCP Address Pool Configuration

Navigate to **Services** → **DHCP**. Fill in the required fields, then click **Add** followed by **Save**.

3. Example DHCP Pool

Parameter	Value
Pool Name	`dotONEnetwork`
Default Gateway	`192.168.1.1`
Start IP Address	`192.168.1.3`
Maximum Number of Users	20

Router Configuration (DHCP Relay)

```
! Engineering Gateway
interface g0/0
 ip address 192.28.1.1 255.255.255.192   ! Gateway IP
 ip helper-address 192.28.1.60           ! DHCP Server
 no shutdown

! Marketing Gateway
interface g0/1
 ip address 192.28.1.161 255.255.255.240 ! Gateway IP
 ip helper-address 192.28.1.60           ! DHCP Server
 no shutdown
```

Configure DHCP on Router

Goal: Use the router itself as the DHCP server for multiple internal subnets.

```
! Configure DHCP pool for dotONEnetwork
R1(config)# ip dhcp pool dotONEnetwork
R1(dhcp-config)# default-router 192.168.1.1   ! Gateway for subnet
R1(dhcp-config)# network 192.168.1.0 255.255.255.0
R1(dhcp-config)# exit
```

NAT Configuration

Preview: Configure NAT on Router 1 to allow internal networks to access the Internet. *GigabitEthernet0/1* is configured as the **outside** interface, while internal LAN and serial interfaces are configured as **inside**. Both **static** and **dynamic** NAT are used.

```
! Internet-facing interface (OUTSIDE)
interface g0/1
 ip address 110.28.29.1 255.255.255.0
 ip nat outside                 ! Mark as NAT outside
 no shutdown

! Static NAT (Engineering PC)
ip nat inside source static 192.28.1.10 110.28.29.1
! Map internal PC to public IP

! Dynamic NAT configuration
access-list 1 permit 192.28.1.0 0.0.0.255 ! Allow internal subnet
```

```
ip nat pool MYPOOL 110.28.29.2 110.28.29.7 netmask 255.255.255.248
! Public IP pool
ip nat inside source list 1 pool MYPOOL ! Enable dynamic NAT

! Inside interfaces
interface g0/0
 ip nat inside                  ! Internal LAN

interface g0/2
 ip nat inside                  ! Internal LAN

interface s0/3/0
 ip nat inside                  ! Internal WAN
```

Configuration C: PAT (NAT Overload)

Goal: Allow multiple internal hosts to access the Internet using a single public IP address on the outgoing interface.

```
! Step 1: Define internal traffic (ACL)
access-list 1 permit 192.168.10.0 0.0.0.255
! Internal subnet allowed for NAT

! Step 2: Enable PAT (NAT Overload)
ip nat inside source list 1 interface fa0/1 overload
! Use outside interface IP for all users
```

OSPF Configuration on Router

Goal: Enable OSPF routing for multiple connected subnets on Router 2.

```
! Enable OSPF process 1
router ospf 1
! Advertise connected networks in area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.31.0 0.0.0.255 area 0
network 10.0.2.0 0.0.0.3 area 0
exit
```

Core Router – Full Configuration (Redistribution Hub)

Goal: Configure the core router with multiple routing protocols and redistribute routes between RIP, OSPF, and static routes.

```
! Enable and enter configuration mode
enable
configure terminal

! -------- Interfaces --------
interface g0/0
 ip address 200.200.20.1 255.255.255.252
```

```
 no shutdown
exit

interface g0/1
 ip address 200.200.20.5 255.255.255.252
 no shutdown
exit

interface g0/2
 ip address 200.200.20.9 255.255.255.252
 no shutdown
exit

! -------- Static route for Router1 LAN --------
ip route 10.10.10.0 255.255.255.0 200.200.20.6  ! Router1 uses static
   routing

! -------- RIP (towards Router0) --------
router rip
 version 2
 no auto-summary
 network 200.200.20.0



 ! Redistribute OSPF and static routes into RIP
 redistribute ospf 1 metric 1
 redistribute static metric 1
exit

! -------- OSPF (towards Router2 + all /30s) --------
router ospf 1
 router-id 1.1.1.1
 network 200.200.20.0 0.0.0.3 area 0
 network 200.200.20.4 0.0.0.3 area 0
 network 200.200.20.8 0.0.0.3 area 0


 ! Redistribute RIP and static routes into OSPF
 redistribute rip metric 10 subnets
 redistribute static metric 10 subnets
exit

! Save configuration
copy running-config startup-config
```

Redistribution Commands – Core Router

Purpose: Allow different routing protocols to share routes across the network.

```
! -------- Redistribute RIP and Static into OSPF --------
router ospf 1
 redistribute rip metric 1 subnets     ! Import RIP routes
```

```
 redistribute static metric 1 subnets  ! Import static routes
exit

! -------- Redistribute OSPF and Static into RIP --------
router rip
 redistribute ospf 1 metric 1          ! Import OSPF routes
 redistribute static metric 1          ! Import static routes
exit
```

Static Routing on R3

Goal: Configure static routes to reach the local LAN with primary and backup paths.

```
! Directly connected static route
R2(config)# ip route 192.168.1.0 255.255.255.0 s0/1/0
                ! Uses directly connected interface as next hop

! Floating static route (backup)
R2(config)# ip route 192.168.1.0 255.255.255.0 192.168.0.2 5
                ! Uses next-hop IP with administrative distance 5
```

Verification Commands

Goal: Verify interface status, routing tables, and connectivity on routers and end devices.

```
! -------- On Routers --------
show ip interface brief        ! Check interface status and IP assignments
show ip route                  ! Display routing table
show ip route rip              ! Display RIP routes
show ip route ospf             ! Display OSPF routes
show ip ospf neighbor          ! Check OSPF neighbors
show ip protocols              ! Verify routing protocols

! -------- From PCs / End Devices --------
ping 10.10.10.10
ping 192.168.0.10              ! Test connectivity to another LAN or VLAN
```

Step 1: Define VLANs on Switches

Goal: Define VLANs on all switches for different departments.

```
! -------- Switch S1 --------
S1(config)# vlan 10
S1(config-vlan)# name Dev
S1(config-vlan)# exit

S1(config)# vlan 20
S1(config-vlan)# name Pro
S1(config-vlan)# exit

S1(config)# vlan 30
S1(config-vlan)# name Admin
```

```
S1(config-vlan)# exit

! Repeat the same VLANs on S2 and S3
```

Step 2: Assign Access Ports to VLANs

Goal: Assign PCs to their respective VLANs on access ports.

```
! Dev PCs (VLAN 10)
S1(config)# interface fa0/1
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 10
S1(config-if)# no shutdown

S1(config)# interface fa0/2
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 10
S1(config-if)# no shutdown

! Pro PCs (VLAN 20)
S1(config)# interface fa0/3
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 20
S1(config-if)# no shutdown

! Admin PCs (VLAN 30)
S1(config)# interface fa0/4
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 30
S1(config-if)# no shutdown

! Repeat for S2 and S3 according to the PCs
```

Step 3: Configure Trunk Links

Goal: Connect switches and router to carry multiple VLANs.

```
!  R o u t e r S1  trunk
S1(config)# interface fa0/24
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk allowed vlan 10,20,30
S1(config-if)# no shutdown

!  S 1 S2  trunk
S1(config)# interface fa0/23
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk allowed vlan 10,20,30
S1(config-if)# no shutdown

!  S 2 S3  trunk
S2(config)# interface fa0/23
S2(config-if)# switchport mode trunk
```

```
S2(config-if)# switchport trunk allowed vlan 10,20,30
S2(config-if)# no shutdown
```

Step 4: Configure Router Subinterfaces (Router-on-a-Stick)

Goal: Enable inter-VLAN routing using router subinterfaces.

```
! VLAN 10 - Dev
Router(config)# interface g0/0.10
Router(config-subif)# encapsulation dot1q 10
Router(config-subif)# ip address 172.17.10.1 255.255.255.0
Router(config-subif)# no shutdown

! VLAN 20 - Pro
Router(config)# interface g0/0.20
Router(config-subif)# encapsulation dot1q 20
Router(config-subif)# ip address 172.17.20.1 255.255.255.0
Router(config-subif)# no shutdown

! VLAN 30 - Admin
Router(config)# interface g0/0.30
Router(config-subif)# encapsulation dot1q 30
Router(config-subif)# ip address 172.17.30.1 255.255.255.0
Router(config-subif)# no shutdown

! Physical interface
Router(config)# interface g0/0
Router(config-if)# no shutdown
```